

Exact $d = 2$ commutativity in a finite strong-operator CPOBC system

To-Marigi
Independent researcher

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Abstract

We give a computer-assisted theorem for a finite, explicitly defined strong-operator model of causal-past-ordered Bell causality (CPOBC). The model is the repository's nonsingular $n \leq 4$ compiler, not the unrestricted quantum sequential-growth algebra. In dimension two, every solution of its literal, printed- Q_{n+1} branch has $[Q_i, Q_j] = 0$ for $1 \leq i < j \leq 4$. The proof removes Q_5 before elimination: 2,564 canonical numerators split exactly into 2,552 Q_5 -free and 12 Q_5 -dependent expressions, and every literal solution satisfies the weaker 2,552-expression shared core. A complete three-stratum cover of the commuting ratios $Q_1^{-1}Q_i$, $i = 2, 3, 4$, reduces the noncommutative alternative to 21 exact rational saturation calculations. All 21 close over $\mathbb{Q}$. Two 21-chart prime-field campaigns agree but are used only as scouts.

As a separate proposition, we verify a scalar-chain open family in which Q_2, Q_3, Q_4 are scalar multiples of Q_1 and the literal Q_5 fiber is GL_2 . That proposition is established by exact evaluation of 1,001 direct reduced-operator relations plus exhaustive, hash-bound compiler-provenance coverage of all 2,564 canonical numerators. It is not a second independent implementation and it does not classify Q_5 over an arbitrary base. Machine-readable certificates, budgets, and negative claim boundaries are included.

1 Problem and claim boundary

Classical sequential growth combines discrete general covariance with a Bell-causality condition on causal-set births [5]. Srivastava and Surya formulate operator-valued quantum Bell-causality orderings and study the transition algebra generated by nonsingular operators Q_n [6]. Their time-ordered and non-time-ordered versions collapse to commutative transition algebras. Their CPOBC ordering instead yields additional relations and leaves the general representation problem open. The displayed $d = 2$ computation in their Appendix starts from a Pauli-proportional ansatz, Eq. (165), and proves that this particular ansatz does not extend to the full algebra. It is not a classification of arbitrary invertible 2 by 2 matrices.

There is also a source-index ambiguity that must remain visible. On PDF page 27, atomisation Eq. (112) is followed by printed Eq. (113) with Q_{n+1} . The Appendix's explicit path calculation, Eq. (163), uses the stage- n generator instead. The repository therefore keeps two source-index branches:

- (i) the *literal branch*, which materialises printed Q_{n+1} ;
- (ii) the *derived branch*, which materialises the Q_n index obtained from the preceding atomisation formula.

No assertion about author intent is made.

The second semantic boundary concerns operator identities. The source paper introduces general covariance on a distinguished initial vector and uses reachable-state reasoning before later atomisation steps employ operator equalities. The repository's `PAPER_STRONG_OPERATOR_PROFILE` materialises the selected GC, MSR, CPOBC, and path relations as strong operator identities. This is an explicit project formalisation and strengthening; it is not described here as a uniform hypothesis of Ref. [6].

Our main statement is consequently finite and conditional on that formalisation.

Theorem 1 (Finite literal-branch $d = 2$ commutativity). *Let $\mathbb{K}$ be an algebraically closed field of characteristic zero. Let*

$$(Q_1, Q_2, Q_3, Q_4, Q_5) \in \text{GL}_2(\mathbb{K})^5$$

satisfy the literal branch of the frozen $n \leq 4$, nonsingular, `PAPER_STRONG_OPERATOR_PROFILE` system released by Final-Theory Bench v0.3.7. Then

$$[Q_i, Q_j] = 0 \quad (1 \leq i < j \leq 4).$$

The conclusion intentionally excludes Q_5 , the transition algebra outside the compiled inventory, weak or occurrence-dependent operator semantics, $d \geq 3$, and the infinite-stage CPOBC algebra.

Remark 1 (The antichain relations alone are insufficient). The conclusion of Theorem 1 does not follow from the gregarious/antichain identities of Ref. [6] alone. Put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Q_n = \sigma_1 D_n, \quad D_n = \text{diag}(a_n, b_n), \quad a_n b_n \neq 0, \quad a_n \neq \pm b_n.$$

The D_n commute and $Q_k^{-1} = D_k^{-1} \sigma_1$, so $Q_n Q_k^{-1} Q_m = \sigma_1 D_n D_k^{-1} D_m$ is symmetric under $n \leftrightarrow m$ and Eq. (120) holds for every admissible triple. Each $Q_m Q_n^{-1} = \sigma_1 D_m D_n^{-1} \sigma_1$ is diagonal, hence Eq. (129) holds for all indices, and Eq. (130) holds likewise. Yet

$$[Q_i, Q_j] = (a_j b_i - a_i b_j) \text{diag}(1, -1),$$

which is nonzero whenever $a_j b_i \neq a_i b_j$. Moreover $\sigma_1 D_n = \frac{a_n + b_n}{2} \sigma_1 + \frac{i(b_n - a_n)}{2} \sigma_2$ is proportional to a single Pauli matrix only when $a_n = \pm b_n$, so under the stated conditions no generator of the family lies inside the ansatz of Eq. (165). The family is *not* a solution of the frozen system; it shows only that the full CPOBC, GC, and MSR inventory is essential to Theorem 1, which is therefore not a restatement of the published antichain identities.

Theorem 1 is also logically independent of Lemma 3.10 of Ref. [6], whose hypothesis is that some fixed Q_k commutes with *every* Q_n , $n \in \mathbb{N}$. Our conclusion is confined to $1 \leq i < j \leq 4$ and says nothing about $[Q_n, Q_k]$ for $n \geq 5$; the finite theorem is consequently not a corollary of that lemma, and it does not activate it. Were a future argument to extend a finite solution to all n with a common central generator, Lemma 3.10 would then apply and give commutativity of the full transition algebra; no such extension is claimed here.

2 Frozen finite system and the Q_5 -free implication

The compiler freezes reduced operator relations and scalarises their 2 by 2 matrix entries after replacing every inverse by its adjugate divided by its determinant. This is a calculation in a

localised commutative coordinate ring; adjugate identities and localisation are standard [7, 4], while exact saturation is a standard ideal operation [1]. These general references do not certify the project-specific relation inventory or any chart result.

For each source-index branch the frozen direct system contains 1,001 matrix relations. The literal scalar system contains 2,564 distinct canonical numerator records. Version 0.3.6 and the v0.3.7 partition certificate give the exact disjoint split

$$2564 = 2552 \text{ } Q_5\text{-independent} + 12 \text{ } Q_5\text{-dependent}.$$

All 191 frozen denominator records are Q_5 -free. At the matrix-relation level, the Q_5 -free subsystem is the 976-relation shared core

$$976 = 700 \text{ CPOBC} + 255 \text{ GC} + 21 \text{ MSR}.$$

The 25 literal path-consistency relations are kept outside that core.

Lemma 1 (Forward implication). *Every solution of the 2,564-numerator literal system satisfies the 2,552-numerator shared core. Therefore, if the shared core forces $[Q_i, Q_j] = 0$ for $i, j \leq 4$, the same commutativity statement holds for every literal solution, independently of Q_5 .*

Proof. The shared core is an exact subset selected by canonical equation IDs, expression IDs, provenance families, and the absence of Q_5 from the symbol inventory. Restricting a solution to a subset of its equations is immediate. Only this direction is used: no equality of solution schemes, reverse implication, minimal-generator claim, or Groebner-basis claim is required. $\square$

This small logical point removes the unsupported premise in the older 49-chart campaign. Version 0.3.5 placed $R_5 = Q_1^{-1}Q_5$ in a simultaneous standard form with R_2, R_3, R_4 , although the compiled $n \leq 4$ inventory did not force $[R_5, R_j] = 0$. Those frozen 49 charts and 225 backend runs remain valid inside that ansatz, but they are not used as a cover of all literal solutions. The v0.3.7 proof never introduces Q_5 or R_5 .

3 Complete ratio strata in dimension two

Set

$$R_i = Q_1^{-1}Q_i, \quad i = 2, 3, 4.$$

The all-stage source relation Eq. (120), specialised only to indices within $1, \dots, 4$, has the form

$$Q_n Q_1^{-1} Q_m = Q_m Q_1^{-1} Q_n.$$

Because Q_1 is invertible, this is equivalent to $[R_n, R_m] = 0$. The relevant instances occur in the frozen shared core. No index-5 instance of the source theorem is imported.

Over an algebraically closed characteristic-zero field, a commuting triple of 2 by 2 matrices has the following exhaustive form, up to simultaneous similarity:

S1: distinct-spectrum pivot. Some R_i has two distinct eigenvalues. It is diagonal, and every matrix commuting with it is diagonal.

S2: common nilpotent direction. There is no distinct-spectrum member, but some R_i is non-scalar. Every member is $\lambda I + \mu N$ for a common nonzero N with $N^2 = 0$.

S3: scalar triple. Every $R_i = \lambda_i I$.

The general simultaneous-similarity theory of 2 by 2 tuples is well known [3, 2]. The particular S1/S2/S3 normal forms, localisation factors, and finite chart split used here are project derivations and are tested against the frozen compiler.

Choosing the first eligible pivot and covering the residual zero/nonzero conditions produces 12 S1 charts and 9 S2 charts. S3 needs no elimination:

$$Q_i = Q_1 R_i = \lambda_i Q_1 \quad (i = 2, 3, 4),$$

so $Q_1, \dots, Q_4$ commute structurally. The S1/S2 charts encode noncommutativity by saturating each independent component of the six commutators $[Q_i, Q_j]$, $1 \leq i < j \leq 4$.

4 Exact chart calculation

For every S1/S2 chart, the 976 direct reduced-operator relations are evaluated over a rational function field and converted to a polynomial ideal over $\mathbb{Q}$. The recorded determinant and chart factors define the open locus. Sequential Sage/Singular saturation then asks whether a nonzero commutator component can survive. A unit ideal is an exact certificate that the corresponding localised branch is empty. Since a unit ideal over $\mathbb{Q}$ remains a unit ideal after extension of scalars, the result applies to the characteristic-zero field $\mathbb{K}$ in Theorem 1.

Table 1: Proof campaign. Prime fields are scouts, not proof fields.

Field	S1 charts	S2 charts	Completed	Role
$\mathbb{Q}$	12	9	21	exact proof
GF(32003)	12	9	21	scout
GF(32009)	12	9	21	scout
Total	36	27	63	

All 21 rational runs have `exit_status=COMPLETED`, `proof_eligible=true`, complete localisation, complete commutator-component coverage, and memory use within the externally owned 8 GiB limit. In S1, nine charts are empty and three have no surviving noncommutative component. In S2, three charts are empty and six have no surviving noncommutative component. The two scout fields reproduce all 21 chart verdicts. They are not promoted to characteristic-zero certificates.

Proof of Theorem 1. Let a literal solution be given. By Lemma 1, its $Q_1, \dots, Q_4$ coordinates satisfy the Q_5 -free shared core. The ratio relations make R_2, R_3, R_4 a commuting triple. The three-stratum lemma places it in S1, S2, or S3 after simultaneous similarity. Every S1 and S2 noncommutative branch has an exact characteristic-zero unit-ideal certificate; S3 is structurally commutative. Hence every commutator among $Q_1, \dots, Q_4$ vanishes. $\square$

5 A scalar-chain Q_5 fiber

The main theorem says nothing about Q_5 . A separate exact identity calculation shows why that omission matters.

Proposition 1 (Literal scalar-chain open family). *Let*

$$Q_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad Q_5 = \begin{pmatrix} e & f \\ g & h \end{pmatrix},$$

$$Q_i = \lambda_i Q_1 \quad (i = 2, 3, 4).$$

On the certificate-defined open set where the 15 recorded irreducible factors are nonzero, all 1,001 frozen literal reduced-operator relations vanish identically. Fourteen factors involve only the base coordinates a, b, c, d and $\lambda_2, \lambda_3, \lambda_4$. The only Q_5 -dependent factor is $fg - eh$. Thus, above every base point in the corresponding 14-factor open set, the literal Q_5 fiber is GL_2 .

The proof has two distinct components, but only one actual evaluation route. First, Sage 10.9 evaluates all 1,001 direct reduced-operator matrix relations over the exact $\mathbb{Q}$ fraction field. In both the literal and derived branches, all 1,001 relation IDs are selected and evaluated, all residual entries are zero, and all 165 reconstructed transition predicates are checked. These runs carry `identity_proof_eligible=true`; the separate ideal-solve flag `proof_eligible` is correctly false.

Second, a static compiler certificate proves that the direct inventory covers the scalar inventory. For each branch:

$$1001 = 979 \text{ represented relations} + 22 \text{ identically-zero path relations.}$$

The 979 represented relations contribute all four matrix entries, giving 3,916 provenance records that cover all 2,564 canonical numerators. The represented relation IDs are exactly the complement of the 22 path IDs. Every source stage, family, branch, original-expression ID, and v0.3.3 residual digest is cross-checked. The compact arena is schema- and SHA-256-bound to the frozen polynomial system, and every referenced target index is in range.

This establishes the proposition without claiming an actual second compact-arena evaluation of 2,564 expressions. The two components share compiler and backend lineage and are not independent implementations. The proposition constructs a noncommutative open family whenever Q_5 is chosen not to commute with Q_1 , but only over the scalar-chain base. It neither classifies Q_5 above an arbitrary literal solution nor produces a representation of the intended infinite CPOBC algebra.

6 Reproducibility and integrity

Table 2 gives the proof-decisive files and selectors. Every backend response is bound to the external budget SHA-256, the full semantic request digest, the selected relation/equation digests, the coefficient field, and the memory limit. The v0.3.7 scope addendum re-opens all 63 Phase-1 certificates, recomputes their hashes and proof gates, reconstructs the two scout maps, and refuses a strong verdict if any chart is absent or ineligible.

The budget file SHA-256 is `f192a37b97e70c84aeae594db75832bdcf632bf00a78009ff45f09b1bcc2e50`. The exact calculation used Sage 10.9 and Singular 4.4.1. The scalar-chain aggregate points only to the final request-bound certificates; superseded and timed-out exploratory certificates are excluded from the release inventory.

Table 2: Machine-readable claim map.

Artifact	Decisive fields
results/v0.3.7_q5_free_partition.json	2564 = 2552 + 12, selected equation/relation digests, no Q_5 in the shared core
results/v0.3.7_q5_free_elimination.json	21 exact QQ runs, 42 scouts, chart coverage, verdict LITERAL_Q1_Q4_COMMUTATIVITY_PROVED
results/v0.3.7_scope_addendum.json	frozen v0.3.4–v0.3.6 hashes, 63 certificate bindings, corrected historical claim scope
results/v0.3.7_general_scalar_chain_fiber.json	1,001 exact identities per branch, 2,564-numerator provenance coverage, 15/14 factor domains, identity proof gate
config/v0.3.7_budget.json	3,600 s per chart, 43,200 s total, 8 GiB; no runtime defaults

7 Limitations and conclusion

The result resolves a precise finite question: within the repository’s nonsingular $n \leq 4$ strong-operator literal system, $d = 2$ forces pairwise commutativity of $Q_1, \dots, Q_4$. It does not settle:

- the intended index in printed Eq. (113);
- weak, fixed-vector, or reachable-state operator semantics;
- a restriction or lifting theorem between the frozen compiler and the full finite- or infinite-stage CPOBC algebra;
- the separately identified general Eq. (139) materialisation gap;
- Q_5 over a general base, dimensions $d \geq 3$, or singular transitions;
- extension of an operator or vector measure to a completed event sigma algebra.

Accordingly, the repository-wide scientific verdict remains `FINAL_THEORY_OPEN`. The finite theorem is stronger and cleaner than the old simultaneous- R_5 ansatz because it uses only a necessary Q_5 -free subsystem. The scalar-chain proposition simultaneously shows that the literal source-index branch can have a free, generically noncommuting Q_5 fiber on a special open base. Keeping these two statements separate is essential: the first is a no-go theorem for $Q_1, \dots, Q_4$ in the frozen literal system, while the second is a bounded construction that exposes the remaining source-index sensitivity.

AI-use disclosure

The author used OpenAI Codex to assist with code review, exact-computation orchestration, test generation, bibliography organisation, and language editing, and Anthropic Claude to assist with research design, the Q_5 -free proof strategy adopted here, independent symbolic cross-checks, and prior-art review. The human author selected the research scope and the resource budgets, reviewed the work, and is responsible for every claim. AI systems are not listed as authors. All proof-bearing claims are tied to exact, inspectable artifacts and tests.

Data and code availability

Source code, frozen inputs, exact certificates, and the paper source are contained in the *universe-theory-lab* repository on the `codex/final-theory-v0.3.7-publication-20260729` branch. No DOI is

claimed in this draft. Release and archival metadata are supplied separately and must not be read as an external publication.

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